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Brief

This assignment is composed of two parts:

1. Study of the configuration language used by a specific network device (IP firewall like nftables or ipfw, or VPN gateway like ip xfrm or OpenVPN).
2. Development of a translator that recognizes the language syntax (or part of it, according to the complexity), and translates it into a JSON or XML document.

Other information:

- Possibility to grant more assignments with different languages.
- Possibility to extend the work with a related thesis work

Questions

Scope & Trade-offs

1. Q: regarding the scope of the assignment, the configuration languages for tools like OpenVPN or nftables are quite extensive. Given the workload, would you prefer us to build a simplified end-to-end version (covering a basic language study and a simple translator for the whole process), or should we focus on a deeper, more comprehensive implementation for just a specific subset of the syntax?

A: simplified end-to-end version

2. Q: IP firewall like nftables or ipfw, or VPN gateway like ip xfrm or OpenVPN, which is better for this project? Or could you recommend a suitable configuration language?

A: nftables

3. Q: Nftables(if choose) has a very large syntax. I plan to focus only on a core subset, like basic IPv4, TCP/UDP filtering rules, and simple actions (accept/drop). Is this scope sufficient for the assignment?*

A: It depends on the verefoo respository requirement

4. Q: Could I use other Parser Generator Library or Framework(Except from Jflex/CUP,such as ANTLR v4)

A: Jflex/CUP

Learning Resources/Schedule

1. Q: To better prepare for the formalization and parsing parts of this project, could you recommend any specific research papers, websites, or reference materials that would be particularly helpful?

A: A.V. Aho, M.S. Lam, R. Sethi, J.D. Ullman, “Compilers, Principles, Techniques, & Tools”, Pearson.

2. Q: When is the best time to start this project (taking into account the pace of the course)? Or complete the relevant section in line with the course schedule?

A: after lab about CUP

3. Q: Just to quickly confirm the logistics: is there a specific hard deadline for the project submission? Also, if I encounter technical blockers during the development phase, is email the best way to reach out for guidance or just ask after every Tuesday's lecture?

A: make a short presentation reservation before each exam

Requirements for Output and Evaluation

1. Output Data Structure: What are the requirements for the generated JSON schema format? Is a generic object representation acceptable?

A: It depends on the verefoo github respository requirement

2. Assessment Criteria: Regarding the final evaluation, will you utilize a set of "blind" test cases, or is the burden of proof on us to provide a suite of test cases demonstrating our tool's reliability? Should I prepare my own set of valid and invalid configuration files to demonstrate error handling?

A: It depends on the verefoo github respository requirement

3. Handle Error: To what extent should a translator handle configuration files containing syntax errors? Should it simply throw an exception and terminate, or should it, like a real compiler, specify the exact line number and type of error? A: like a real compiler, specify the exact line number and type of error**